This is a slightly generalized form of the technique suggested by Bulgac and Kusnezov [58].

Assuming ρ goes to zero sufficiently rapidly at infinity, we have

$$\int \nabla_z \cdot \boldsymbol{G}(z)\rho d\omega = -\int (\nabla_z \rho)\cdot \boldsymbol{G}(z) d\omega = \beta \int \nabla_z H \cdot \boldsymbol{G}(z)\rho d\omega,$$

using integration by parts, therefore, since $\beta^{-1} = \mathrm{k_B T}$,

$$\mathrm{T} = \mathrm{k_B^{-1}} \frac{\langle \nabla_z H \cdot \boldsymbol{G}\rangle}{\langle \nabla \cdot \boldsymbol{G}\rangle},$$

which gives formulas for temperature in terms of phase space averages (compare Chap. 6, Proposition 6.1).

Suppose we let K be a scalar valued function of position, and set

$$\boldsymbol{G}(z) = \begin{bmatrix} \nabla_q K(\boldsymbol{q}) \\ \boldsymbol{0} \end{bmatrix}$$

then we arrive at

$$g(z) = \mu^{-1}[\nabla_q U - \mathrm{k_B T}\nabla_q] \cdot \nabla_q K(\boldsymbol{q}).$$

This is an example of a "configurational thermostat" [193, 211, 319].

A useful fact about thermostats is the following:

Proposition 8.1 (Thermostats are Additive) *Suppose we are given two thermostats,*

$$\dot{z} = \boldsymbol{J}\nabla H + \boldsymbol{\Phi}_1(\boldsymbol{\xi}, z), \qquad \dot{\boldsymbol{\xi}} = \boldsymbol{g}_1(\boldsymbol{\xi}, z),$$

with stationary density $\hat{\rho}_1(\boldsymbol{\xi})\rho_\beta$, and another

$$\dot{z} = \boldsymbol{J}\nabla H + \boldsymbol{\Phi}_2(\boldsymbol{\eta}, z), \qquad \dot{\boldsymbol{\eta}} = \boldsymbol{g}_2(\boldsymbol{\eta}, z),$$

with stationary density $\hat{\rho}_2(\boldsymbol{\eta})\rho_\beta$, then

$$\begin{aligned} \dot{z} &= \boldsymbol{J}\nabla H + \boldsymbol{\Phi}_1(\boldsymbol{\xi}, z) + \boldsymbol{\Phi}_2(\boldsymbol{\eta}, z), \\ \dot{\boldsymbol{\xi}} &= \boldsymbol{g}_1(\boldsymbol{\xi}, z), \\ \dot{\boldsymbol{\eta}} &= \boldsymbol{g}_2(\boldsymbol{\eta}, z) \end{aligned}$$

is another thermostat and has stationary density $\tilde{\rho} = \hat{\rho}_1(\boldsymbol{\xi})\hat{\rho}_2(\boldsymbol{\eta})\rho_\beta$.

The proof is an immediate consequence of the linearity of the divergence operator, since the Liouville operator is constructed from this.

8.2.2 Nosé-Hoover Chains

Let us employ the additivity property to derive "Nosé-Hoover Chains" [251], a popular deterministic thermostat designed to give a more thorough sampling of the phase space than the standard Nosé-Hoover method.

The equations of Nosé-Hoover Chains may be written [251]

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \tag{8.16}$$

$$\dot{\boldsymbol{p}} = -\nabla U(\boldsymbol{q}) - \xi_1 \boldsymbol{p}, \tag{8.17}$$

$$\mu_1 \dot{\xi}_1 = \boldsymbol{p}^T \boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d k_B T} - \mu_1 \xi_2 \xi_1, \tag{8.18}$$

$$\mu_2 \dot{\xi}_2 = \mu_1 \xi_1^2 - \mathrm{k_B T} - \mu_2 \xi_2 \xi_3, \tag{8.19}$$

$$\mu_3 \dot{\xi}_3 = \mu_2 \xi_2^2 - \mathrm{k_B T} - \mu_3 \xi_3 \xi_4, \tag{8.20}$$

$$\vdots$$

$$\mu_k \dot{\xi}_k = \mu_{k-1} \xi_{k-1}^2 - \mathrm{k_B T}. \tag{8.21}$$

By creating a more substantial bath of oscillators, each of which is forced to have the target kinetic energy, the physical system may more closely approximate the canonical distribution.

To see that this is a thermostat, note the equations (8.16)–(8.21) can be split into the following pieces:

$$\begin{bmatrix} \dot{\boldsymbol{q}} \\ \dot{\boldsymbol{p}} \\ \mu_1\dot{\xi}_1 \\ \mu_2\dot{\xi}_2 \\ \vdots \\ \mu_{k-1}\dot{\xi}_{k-1} \\ \mu_k\dot{\xi}_k \end{bmatrix} = \begin{bmatrix} \boldsymbol{M}^{-1}\boldsymbol{p} \\ -\nabla U(\boldsymbol{q}) - \xi_1\boldsymbol{p} \\ \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d k_B T} \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ -\mu_1\xi_1\xi_2 \\ \mu_1\xi_1^2 - \mathrm{k_B T} \\ \vdots \\ 0 \\ 0 \end{bmatrix} + \cdots + \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ 0 \\ 0 \\ \vdots \\ -\mu_{k-1}\xi_{k-1}\xi_k \\ \mu_{k-1}\xi_{k-1}^2 - \mathrm{k_B T} \end{bmatrix}.$$

The first piece of the system corresponds to Nosé-Hoover, which we already know preserves $\rho_\beta \exp(-\beta\mu_1\xi_1^2/2)$. By examination, the distribution $\exp(-\beta\mu_1\xi_1^2/2)$ $\exp(-\beta\mu_2\xi_2^2/2)$ is preserved by the second piece, the next piece preserves